

Brazilian Football Modelling

Calibrated Bayesian Models at Club and Player Levels

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Roadmap

- ① Literature Review
- ② Data
- ③ Models
- ④ Simulation-Based Calibration
- ⑤ Results
- ⑥ Conclusion

Literature Review

- ▶ **Bradley-Terry models:** proposed by [Bradley e Terry \(1952\)](#)
 - ▶ Extensions: ties on [Davidson \(1970\)](#) and home effect on [Davidson e Beaver \(1977\)](#);
- ▶ **Poisson models:**
 - ▶ Simple Poisson model on [Maher \(1982\)](#), extended with home effect on [Dixon e Coles \(1997\)](#) and [Lee \(1997\)](#);
 - ▶ [Karlis e Ntzoufras \(2003\)](#) uses a bivariate Poisson;
- ▶ **Player-level:** Elo-based player ratings proposed by [Wolf, Schmitt e Schuller \(2020\)](#);
- ▶ **Validation:** Simulation-Based Calibration, formally introduced by [Cook, Gelman e Rubin \(2006\)](#) and [Talts et al. \(2020\)](#).

Data

Dataset Overview

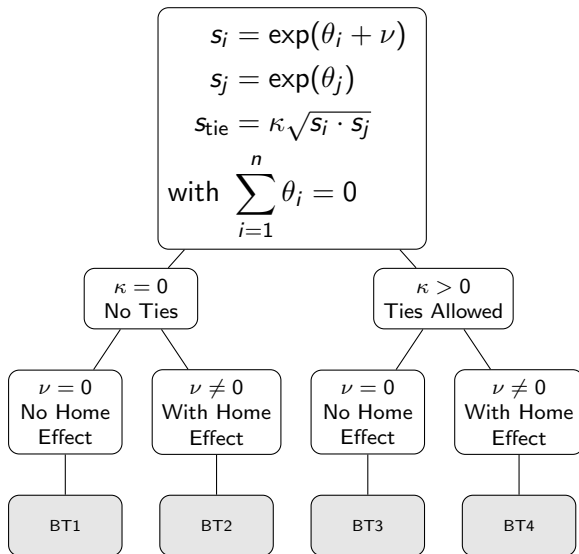
- ▶ Extracted from CBF's PDF match reports;
- ▶ Contains 16,000+ players and 2,000+ matches;
- ▶ 368 clubs and 5 competitions between 2013 and 2025;

	<i>Série A</i>	<i>Série B</i>	<i>Série C</i>	<i>Série D</i>	<i>CdB</i>
Total clubs	37	66	83	297	294
Total players	3,467	4,657	4,738	11,141	9,782

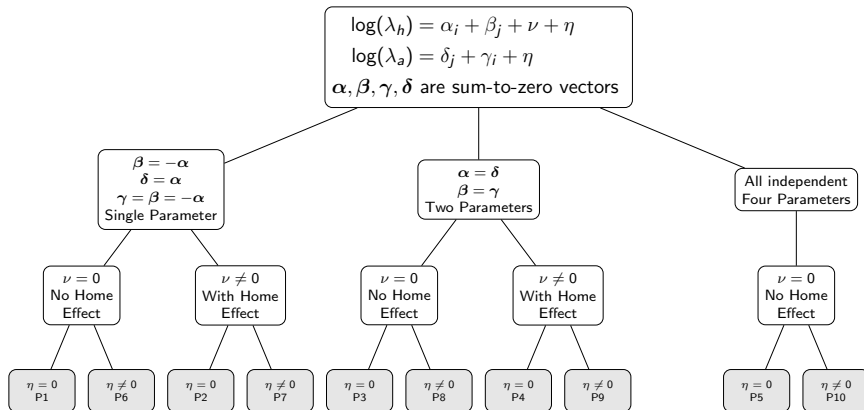
Unique clubs and players across all seasons per competition.

Models

Bradley-Terry – Model Taxonomy



Poisson – Model Taxonomy



Player-Level Extension

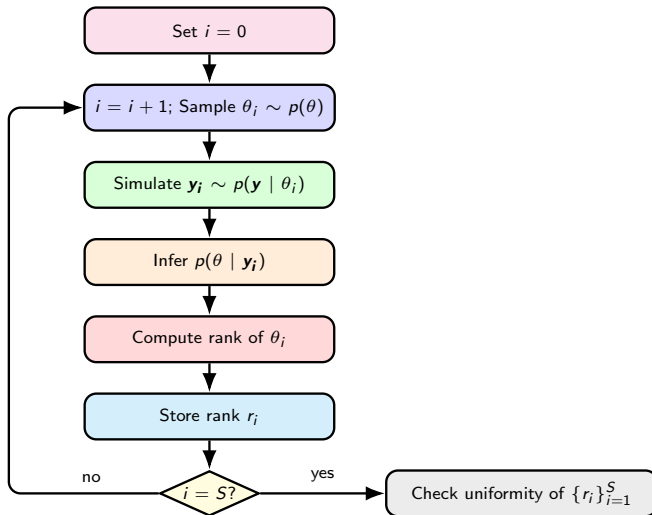
We replace the team skill with aggregated player skills:

$$s_{\text{team}} = \sum_{p \in \mathcal{P}} \exp(\theta_p) \cdot \frac{m_p}{90}$$

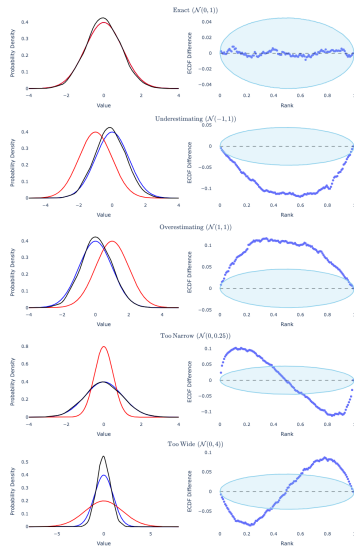
- ▶ θ_p is the individual log-skill, with $\sum_p \theta_p = 0$;
- ▶ we use a synthetic player with an estimable parameter $\theta_{\text{synthetic}}$ to model the impact of red cards.

Simulation-Based Calibration

SBC – Workflow



SBC – Diagnostics



- **Flag:** ≥ 1 ECDF point outside the envelope for a parameter;
- **Points Out:** total violations across all rank values in that model;
- Expected flagging rate under correct calibration: $\approx 5\%$.

Results

Scoring Rules

Let $\mathbf{z}_t$ be the outcome and $\mathbf{p}_t$ the predicted probabilities.

► **Brier Score:**

$$\text{BS} = \frac{1}{n} \sum_{t=1}^n \sum_j (z_{j,t} - p_{j,t})^2$$

► **Log Score:**

$$\text{LS} = -\frac{1}{n} \sum_{t=1}^n \sum_j z_{j,t} \log(p_{j,t})$$

► **Ranked Probability Score:**

$$\text{RPS} = \frac{1}{2n} \sum_{t=1}^n \sum_{k=1}^2 \left(\sum_{j=1}^k (z_{j,t} - p_{j,t}) \right)^2$$

Club-Level Model Comparison

Model	BS	RPS	LS	Description
Naive 1	0.6667	0.2359	1.0986	Equal probabilities
Naive 2	0.6373	0.2243	1.0570	Historical proportions
BT3	0.6494	0.2293	1.0768	Bradley-Terry with ties
BT4	0.6508	0.2296	1.0800	BT3 + home effect
P1	0.6497	0.2287	1.0741	Simple Poisson
P2	0.6256	0.2180	1.0402	P1 + home effect
P3	0.6558	0.2324	1.0860	Attack/defence
P4	0.6356	0.2230	1.0550	P3 + home effect
P5	0.6834	0.2460	1.1255	Home/away split
P6	0.6495	0.2286	1.0738	P1 + correlation
P7	0.6237	0.2172	1.0378	P2 + correlation
P8	0.6548	0.2318	1.0840	P3 + correlation
P9	0.6334	0.2218	1.0526	P4 + correlation
P10	0.6835	0.2460	1.1263	P5 + correlation

Predictive performance on *Brasileirão* 2020–2025. Lower values are better; best model is in bold.

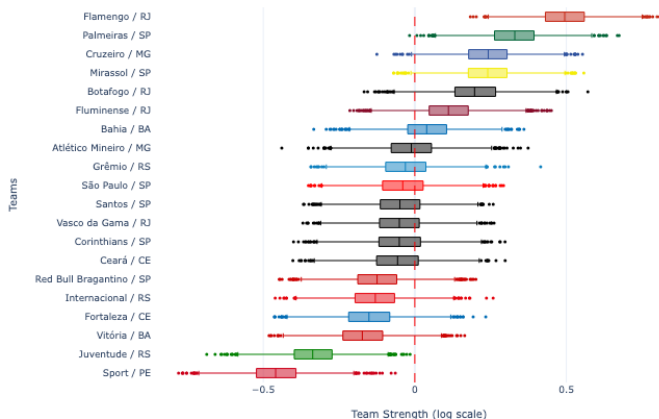
Match-Level Parameters by Season (P7)

Season	ν (95% CI)	$\exp(\nu)$	η (95% CI)	$\exp(\eta)$
2020	0.273 (0.15, 0.40)	1.31	0.031 (−0.07, 0.13)	1.03
2021	0.296 (0.16, 0.43)	1.34	−0.096 (−0.20, 0.00)	0.91
2022	0.365 (0.23, 0.50)	1.44	−0.075 (−0.18, 0.04)	0.93
2023	0.281 (0.14, 0.40)	1.32	0.028 (−0.07, 0.13)	1.03
2024	0.310 (0.19, 0.44)	1.36	−0.003 (−0.11, 0.09)	1.00
2025	0.425 (0.29, 0.55)	1.53	−0.062 (−0.16, 0.04)	0.94

Summary of the posterior distributions of the home effect parameter (ν) and the correlation parameter (η) across seasons.

Team Strengths – 2025

Team Strengths - Serie A 2025 (after 380 games)



Posterior distributions of the team strength parameter (log scale) for each club, ordered by posterior median strength.

Player Rankings (2020–2025)

Top 20					
#	Player	Club	Med	95% CI	WGD
1	Weverton	PAL	0.132	[−0.07, 0.33]	153.5
2	Arrascaeta	FLA	0.093	[−0.10, 0.30]	106.7
3	F. Bruno	FLA	0.083	[−0.12, 0.28]	94.2
4	G. Gómez	PAL	0.083	[−0.12, 0.29]	105.5
5	Murilo	PAL	0.082	[−0.12, 0.28]	96.4
6	J. Piquerez	PAL	0.079	[−0.12, 0.28]	99.9
7	Marcos Rocha	PAL	0.076	[−0.12, 0.27]	91.1
8	R. Veiga	PAL	0.076	[−0.13, 0.28]	93.0
9	Léo Pereira	FLA	0.067	[−0.13, 0.27]	84.0
10	Zé Rafael	PAL	0.064	[−0.14, 0.26]	82.3
11	A. Rossi	FLA	0.063	[−0.14, 0.26]	74.0
12	Éverson	CAM	0.060	[−0.14, 0.26]	72.0
13	Renê	INT	0.058	[−0.14, 0.26]	61.0
14	Rony	PAL	0.056	[−0.14, 0.25]	73.1
15	Léo Ortiz	RBB	0.055	[−0.15, 0.25]	64.7
16	Pedro	FLA	0.052	[−0.15, 0.25]	65.9
17	B. Henrique	FLA	0.052	[−0.15, 0.25]	64.7
18	Ayrton Lucas	FLA	0.052	[−0.15, 0.25]	61.6
19	Gerson	FLA	0.050	[−0.15, 0.25]	63.5
20	Cuesta	INT	0.050	[−0.15, 0.25]	55.3

Bottom 20					
#	Player	Club	Med	95% CI	WGD
1956	Victor Luis	BOT	−0.026	[−0.22, 0.17]	−32.7
1957	Lucas	AME	−0.026	[−0.22, 0.17]	−34.7
1958	Rodrigo Alves	JUV	−0.026	[−0.22, 0.16]	−34.4
1959	Pedro Raul	CEA	−0.026	[−0.22, 0.16]	−37.9
1960	Victor Hugo	BAH	−0.027	[−0.22, 0.16]	−36.8
1961	Shaylon	ATG	−0.028	[−0.22, 0.17]	−36.6
1962	R. Thyere	SPO	−0.028	[−0.22, 0.17]	−37.2
1963	Lucas Halter	GOI	−0.029	[−0.22, 0.16]	−39.1
1964	C. Rivera	SPO	−0.029	[−0.22, 0.16]	−39.5
1965	Gilberto	BAH	−0.030	[−0.22, 0.16]	−38.8
1966	Matheus	CUI	−0.030	[−0.22, 0.16]	−37.9
1967	G. Baralhas	ATG	−0.031	[−0.23, 0.16]	−44.6
1968	Éder	ATG	−0.034	[−0.22, 0.16]	−46.0
1969	Tadeu	GOI	−0.035	[−0.22, 0.15]	−48.0
1970	Matheusinho	SPO	−0.036	[−0.23, 0.16]	−41.3
1971	I. Pitta	CUI	−0.037	[−0.23, 0.15]	−48.8
1972	Kevin	AVA	−0.037	[−0.23, 0.15]	−47.6
1973	Lucas Lima	SPO	−0.044	[−0.24, 0.15]	−54.0
1974	Jadson	JUV	−0.050	[−0.24, 0.14]	−66.4
1975	G. Vasconcelos	COR	−0.051	[−0.24, 0.14]	−71.4

Conclusion

Conclusion

- ▶ Model P7 (single skill + home + correlation) performs best;
- ▶ Split attack/defence by home/away does not improve performance;
- ▶ A correlation parameter between goals is not statistically significant;
- ▶ Home effect is significant: home team expected goals between 1.31 and 1.53 times when playing away;
- ▶ Player-level rankings reflect team context ($r \approx 0.99$ with WGD).

- ▶ Player-specific event data for finer attribution;
- ▶ Dynamic models: team/player abilities evolving over time;
- ▶ Contextual covariates: fatigue, travel, match importance;
- ▶ All code is available at <https://github.com/BrazilianFootball>.

Thank you!